ELECTRICAL CIRCUIT AND MACHINES

Unit	Contents
Unit-1	**Basic Concepts of Electrical Circuits** 1.1 Concept of electric charge, current, potential difference, power, and energy. Ohm's law and its 1.2 Resistance, capacitance, and inductance—definitions, units, and their effects in an electrical circuit. 1.3 Series and parallel circuits, Kirchhoff's laws, and their applications in circuit analysis (simple problems). 1.4 Star-Delta transformation (derivation not required).
Unit-2	**AC Fundamentals and Network Theorems** 2.1 Sinusoidal waveforms, RMS and average values, form factor, and peak factor. 2.2 Concept of reactance and impedance, phase and phase difference, power in AC circuits (active, reactive, and apparent power), power factor. 2.3 Series and parallel AC circuits, resonance in AC circuits, bandwidth, and Q-factor. 2.4 Network theorems—Thevenin's, Norton's, Superposition, Maximum Power Transfer Theorem (statement, explanation, and numerical problems).
Unit-3	**DC Machines** 3.1 Working principle, construction, and types of DC machines. 3.2 EMF equation, characteristics of DC generators and motors, torque equation. 3.3 Starting methods of DC motors, speed control of DC motors. 3.4 Applications of DC machines.
Unit-4	**Transformers** 4.1 Working principle, construction, and types of transformers. 4.2 EMF equation, transformation ratio, efficiency, and voltage regulation. 4.3 Losses in a transformer, OC and SC tests, and condition for maximum efficiency. 4.4 Auto-transformer and its applications.
Unit-5	**Three-Phase Induction Motors** 5.1 Working principle, construction, and types of three-phase induction motors. 5.2 Slip and its significance, torque equation (derivation not required). 5.3 Methods of starting and speed control of induction motors. 5.4 Applications of three-phase induction motors.
Unit-6	**Single-Phase Motors and Special Machines** 6.1 Types of single-phase motors—capacitor start, capacitor run, shaded pole motors, universal motors. 6.2 Stepper motors and servo motors—working principle and applications.
Unit-7	**Synchronous Machines** 7.1 Construction, working principle, and types of alternators. 7.2 EMF equation and voltage regulation of alternators (numerical problems). 7.3 Working principle of synchronous motors, torque-angle characteristics, V and inverted V curves.

ELECTRICAL CIRCUIT AND MACHINES (PRACTICAL)

Unit	Experiment Title
Unit-1	**Basic Electrical Circuits** 1.1 Measurement of voltage, current, and resistance using a multimeter. 1.2 Verification of Ohm's Law and Kirchhoff's Voltage and Current Laws. 1.3 Measurement of resistance in series and parallel circuits. 1.4 Verification of Thevenin's and Norton's Theorems. 1.5 Verification of Superposition and Maximum Power Transfer Theorem. 1.6 Star-Delta transformation verification.
Unit-2	**AC Circuits and Power Measurement** 2.1 Measurement of AC voltage, current, and power using a single-phase wattmeter. 2.2 Measurement of power factor and improvement using a capacitor. 2.3 Resonance in series and parallel RLC circuits. 2.4 Measurement of impedance and phase angle in RL, RC, and RLC circuits. 2.5 Study and testing of three-phase power measurement using two-wattmeter method. 2.6 Measurement of active, reactive, and apparent power in AC circuits.
Unit-3	**DC Machines** 3.1 Study of construction and working of DC generator and motor. 3.2 Measurement of generated EMF in a separately excited DC generator. 3.3 Load test on DC shunt motor to determine efficiency and speed regulation. 3.4 Speed control of DC motor by armature voltage and field current control methods. 3.5 Study and testing of DC motor starters (three-point and four-point starters). 3.6 Reversal of rotation in DC motors and its applications.
Unit-4	**Transformers** 4.1 Identification of transformer parts and testing of windings using a multimeter. 4.2 OC and SC tests on a single-phase transformer to determine efficiency. 4.3 Load test on a transformer to find efficiency and voltage regulation. 4.4 Polarity test and parallel operation of transformers. 4.5 Study of auto-transformer and its applications. 4.6 Measurement of three-phase transformer connections (Star-Star, Delta-Delta, Star-Delta, Delta-Star).
Unit-5	**Three-Phase Induction Motors** 5.1 Identification and study of different parts of an induction motor. 5.2 Load test on a three-phase induction motor to determine efficiency. 5.3 Study of various starting methods (DOL, Star-Delta, Auto-transformer starter). 5.4 Speed control of three-phase induction motor using V/F method. 5.5 Measurement of slip and torque characteristics of an induction motor.
Unit-6	**Single-Phase Motors and Special Machines** 6.1 Study of single-phase capacitor start and capacitor run motor. 6.2 Load test on a universal motor and demonstration of its applications. 6.3 Study of stepper motor and servo motor control methods.
Unit-7	**Synchronous Machines** 7.1 Study of alternator parts and determination of EMF equation. 7.2 Voltage regulation measurement of an alternator using synchronous impedance method. 7.3 V and inverted V curves of a synchronous motor.